

Feature Inventory

everything built since the fork

The systems, content, and tooling layered on top of Berryhunter's survival foundation to transform it into **Aura** — a top-down aura-combat MMO-lite. Each feature is described from both the in-game and the technical side.

29 FEATURES	44 PLAYER SKILLS	44 MOB TYPES	10 COMBO RECIPES
11 FACTIONS	2 ZONES	1 WORLD BOSS	21 EFFECT TYPES

Core skill / aura engine

1 Skill component framework — the spine of everything

GAME Every player and mob carries a spellbook and a loadout: several equipped auras but **exactly one active at a time** (a field around you), always-on passives, and cooldown abilities. Switch auras mid-fight, spend/refund points, level skills.

TECH `skills/component.go` — `SkillComponent` (aura/passive/cooldown slots, active slot, spellbook, derived stats). One resizable circular sensor per entity, not one per skill.

2 Data-driven skill definitions + effect vocabulary

GAME Each skill is composed of effects — damage, heal, slow, resist, shield, damage-over-time, heal-over-time, summon, taunt, light, recall, dash, revive, haste, stat buff — with per-level scaling on damage / heal / radius / target-count / tick-rate.

TECH `skills/definition.go` — 21 `EffectType`s parsed from `api/skills/*.json` with strict per-type key allowlists (hard-fail on typos). Three categories: `active_aura` / `passive` / `cooldown`.

3 Skill application system

GAME Makes auras pulse on their tick cadence, fires cooldowns (player input, or auto for mobs), rolls crits/variance, applies no-friendly-fire and target caps.

TECH `sys/skills.go` (the big one). Per-tick sensor resize, per-effect intervals, player↔mob damage paths, cast-time wind-up/interrupt, `mayHarm` hostility gate, faction-relative target eligibility.

4 Capped, no-aim targeting

GAME No manual aiming — each aura picks its own targets by rule (nearest / lowest-% health / all), capped to N. Positioning decides who gets hit.

TECH `sys/targeting.go` — range filter → selector sort → take N. `selector` / `maxTargets` (+per-level) on damage/heal/instant effects.

5 Transient buff / debuff store

GAME Burns keep ticking after you leave the aura, shields absorb hits, slows / resists / haste linger.

TECH `skills/buffs.go` — one generic `Bufs` container per entity, keyed by source skill; `AbsorbShield`, `ResistMultiplier`, `SlowFraction`, `TickRateFactor`.

6 HP, resistances & damage tags, stat variance

GAME Absolute HP pools; mobs resist / are immune to specific damage types (string tags, incl. `"*"` wildcard); harvest & gate obstacles only take damage from the "right" aura; per-hit and spawn-HP variance.

TECH Resist maps in `skills/resist.go`, gated damage tags, spawn-HP bands (the `plan-item11` work).

Progression & unlocks

7 Spellbook, skill points, free respec

GAME Discover skills (never lost on death), spend points earned per level to raise levels, freely respec down to discovery level.

TECH `skills/component.go` — spellbook, point accounting, raise/lower level; skills survive death via carried component in `sys/state.go`.

8 Five unlock sources

GAME Skills come from **milestone levels**, **mob kill-drops** (chance, shared with recent healers), **NPC teaching**, **recipes**, and (scoped) character sacrifice.

TECH `skills/milestones.go`; `MobUnlock` in `items/mobs/definitions.go`; `sys/npc.go`; `skills/recipe*.go`.

9 Aura combinations / recipes

GAME Secret, undocumented recipes — hold a set of skills at required levels and a capstone auto-unlocks (Vanguard trio → Warbanner). Community-discovered.

TECH `skills/recipe.go` + `recipe_apply.go` — cascading, idempotent threshold check on every discovery / level-up.

10 XP for all participants

GAME Every damager (and their recent healers) gets full XP on a kill — no groups, no split, no grief.

TECH `model/mob/mob.go` `rewardPlayer` — participant + recent-healer tracking, then runs the recipe cascade.

Mobs & AI

11 Mob AI depth

GAME Threat table (per-player aggro, taunt / detaunt), state-dependent leash + walk-home, out-of-combat regen, flee-when-low, auras-off-until-aggroed. Idle archetypes: stationary / local wander / route patrol with evade-return, plus obstacle steering around props.

TECH `model/mob/mob.go`, `patrol.go`, `steering.go` — threat ops, invuln/flee overrides, TTL, buff appliers.

12 Faction system

GAME Players + summons are "aligned"; content can declare factions friendly-to-players (never harmable, still fight other factions), plus mob-vs-mob hostility.

TECH `model/faction.go` — `HostilityGate.MayHarm`, `PlayerFriendly`; enforced in `sys/skills.go`.

13 Summons / companions

GAME Cooldowns spawn owned, player-aligned mobs: stationary totems or following companions that defend-then-assist and expire on a timer; owner-attributed XP.

TECH `sys/skills.go` `spawnSummon`; `model/mob/companion.go` — follow ring, tether, owner combat-signal acquisition, teleport catch-up.

14 Support mob (seek-healer)

GAME Friendly healer mob seeks the most-wounded ally, chases it; heal aura toggles on only while healing.

TECH `model/mob/healer.go` — reuses the aggro machinery with allies instead of enemies.

15 Encounter controller + Orc Warlord world boss

GAME First designed boss: invulnerable while either Warbanner totem stands; grunt waves at 66 / 33%; banners replant; wiping resets it; killing it broadcasts the credited players server-wide, empties the arena, and respawns him ~5 min later.

TECH `encounter/system.go` (ECS lifecycle hooks, spawn/despawn/announce); `encounter/warlord.go` (script); positions from zone anchors.

World & atmosphere

16 Authored zone system

GAME The procedural world is replaced by hand-authored zones: bounds, terrain, props, spawns, campfires, dark areas, teaching NPCs, and named encounter anchors; `-zone` selectable.

TECH `world/zone.go` — `LoadZoneFS` with strict validation; anchors, campfires, dark areas, spawn archetypes, prop registry.

17 NPC teaching / lore

GAME Peaceful, unattackable NPCs teach ordered skills on approach (level-gated, "too low" line) or speak lore / sign-post lines.

TECH `sys/npc.go` — rising-edge proximity; `onApproach` grants + recipe cascade; `speak` reuses the chat wire.

18 Campfire recovery, respawn, revive & recall

GAME Campfires heal a % of max HP and, after dwelling, become your respawn point; fresh players spawn at a starting campfire; Recall teleports you back; Revive raises a downed player at their corpse.

TECH `sys/state.go` — campfire anchors, dwell tracking, default spawn, `tryRespawn`, `ReviveAtCorpse`, `AnchorOf`.

19 Light & darkness

GAME Authored dark-area circles independent of day/night; a light aura or Torch passive punches a lit hole — a real slot trade-off (light vs. combat aura). Purely presentational.

TECH `light_aura` effect + `light_radius` wire (max over active aura and passives); client `DarknessOverlay.ts`.

20 Chat / server broadcast

GAME Proximity speech bubbles plus server-wide system banners (boss beats, `ANNOUNCE`).

TECH `sys/chat/system.go` — `Broadcast` (sender id 0 = system banner) + proximity say via viewport.

Client / UI

21 Spellbook UI

GAME Scrollable Auras / Passives / Cooldowns panel, level badges with +/- spend buttons gated by skill points, gold glow on genuine unlocks (level-ups & baseline never false-trigger).

TECH `HUD.ts` setup/update + `SimpleBar` scroll; diff logic against known spellbook ids/levels.

22 Ability bar & loadout

GAME Aura slots (hotkeys 1-4, one active / toggle-to-Nothing), passive slots, cooldown slots (Q/E) with remaining-seconds readout; loadout editing locked in combat with feedback.

TECH `HUD.ts`, `Controls.ts`, `InputMessage.ts` — server-authoritative with optimistic-highlight reconciliation.

23 Aura ring rendering

GAME Translucent field around the caster: red damage / green support / **both rings** for hybrids (Paladin / Vanguard / Warbanner); an edge glow pulses in time with each tick and speeds up under haste. Per-tick hit VFX (slash vs. fire) + floating damage / heal / XP numbers.

TECH `Character.ts`, `AuraTickIndicator.ts` — ring selection, radius, tick-phase glow.

24 AlertBanner

GAME Upper-third single-slot queued banner for server announcements + gold new-skill unlock notices.

TECH `alert-banner/AlertBanner.ts` — `queue` + `show(text, kind)`; fed from `EntityMessage` id 0.

25 Portrait rotation freeze

GAME Avatars are fixed-facing portraits that never spin to the pointer (local player included).

TECH `Character.ts` — `setRotation` forces downward facing; held items still transform.

26 Zone editor dev panel

GAME In-game placement / editing of props, spawns, campfires, dark areas, NPCs, and anchors, with hit-testing and a registry-validated teaching-skill dropdown.

TECH `zone-editor/ZoneEditor.ts` + `ZoneModel.ts` — per-kind place/update/remove/hitTest, marker drawing.

27 Mob roster art

GAME ~40 new `EntityTypes` / SVGs: wildlife, kobolds (melee / ranged), spiders + venom, bandits (ranged / healer / elite / pyromancer), army & orc faction incl. the Warlord boss, player summons, and hazards (PoisonPool, Rockfall, Bramble, SpikeBarricade, Brazier, Campfire, Turnip).

TECH `Mobs.ts` (~40 classes, each registers an SVG); NPCs ride the Resource path in `Resources.ts`.

Tooling

28 Dev cheats

GAME Token-gated testing commands: `GOD`, `WARP x y`, `SPEED [factor|off]`, `XP <n>`, `SKILL <name>`, `KILL`, `DAMAGE <pct>`, `THREAT [id]`, `ANNOUNCE <text>`.

TECH `sys/cmd/cmd.go` `CommandSystem` — static map + space/announcer-bound closures.

29 Simulation harness

GAME Headless balancing explorer that drives the real ECS: TTK / TTD, kills-per-level, 1-vs-N matrix, stand-still-bot facetank thresholds, plus a `-serve` web UI.

TECH `cmd/simharness`.

Two things worth flagging as NOT built despite appearances. Aura line-of-sight was deliberately cut — auras pass through walls; walls only block movement (LoS in the client is the soft light/darkness hole model, not raycast occlusion). **Skills.ts is hand-synced tech debt** — a skill's name / category / max-level is duplicated client-side and scheduled to become a server catalog message.

Roster snapshot (working tree): ~44 player skills (20 auras / 18 cooldowns / 6 passives) · ~33 mob skills · ~44 mobs · 11 factions · 10 combination recipes · 2 zones + 1 boss encounter. All numbers are placeholders until marked final.